

WRRmod2026 Preliminary Program 29/01/2026

Sunday June 14, 2026			
Activity	Time	Details	Speakers
YWP Workshop	08:30 – 10:00	Presentations and Discussion	YWP Workshop Co-Chairs:
	10:00 – 10:30	Break	
	10:30 – 12:00	Presentations and Discussion	
	12:00 – 1:30	Lunch	
	1:30 – 3:00	Presentations and Discussion	
	3:00 – 3:30	Break	
	3:30 – 5:00	Presentations and Discussion	
YWP Social	5:00-7:00	YWP Social Event	
Dinner	7:30-9:00	Dinner	

Monday June 15, 2026			
Activity	Time	Details	Speakers
Parallel Half and Full-Day Workshops See list of workshop titles below	08:30 – 10:00	Presentations and Discussion	See workshop descriptions.
	10:00 – 10:30	Break	
	10:30 – 12:00	Presentations and Discussion	
	12:00 – 1:30	Lunch	
	1:30 – 3:00	Presentations and Discussion	
	3:00 – 3:30	Break	
	3:30 – 5:00	Presentations and Discussion	
Dinner	7:30-9:00	Dinner	

WS	Length	Workshop Title
1	Morning	Mechanistic modelling for mainstream advanced short-cut nitrogen removal with partial denitrification anammox (PdNA) Sara Arabi¹, Haydee De Clippeleir², Imre Takacs³, Tanush Wadhawan³, Ahmed Al-Omari⁴ ¹Stantec, Denver, ²DC Water, ³Dynamita, ⁴Brown and Caldwell
2	Afternoon	Modelling and implementation of low dissolved oxygen BNR systems: from theory to practice Jose Jimenez¹, Peter Dold², Juan Baeza³, Joshua Boltz⁴, Leon Downing⁵, Gillian Burger², Manel Garrido¹ ¹ Brown and Caldwell, ² EnviroSim Associates Ltd, ³ Universitat Autònoma de Barcelona (UAB), ⁴ Woodard and Curran, ⁵ Black and Veatch
3	Morning	How to shape good modelling practice for hybrid mechanistic/data-driven modelling HC Croll¹, L Verhaeghe^{2,3}, E Arispe⁵, E Torfs², P Dunlap⁶, H Stewart⁷, S Daneshgar⁴, A Froemelt⁸, JJ Zhu⁹, MY Schneider^{2,10} ¹ HDR Inc., Cincinnati, ² modelEAU, Université Laval, ³ BIOVISM, Ghent University, ⁴ BIOMATH, Ghent University, ⁵ CDM Smith, ⁶ Black & Veatch, ⁷ Jacobs, ⁸ Eawag, ⁹ Princeton University, Princeton, ¹⁰ DTU Sustain, Technical University of Denmark
4	Afternoon	CFD For Digital Twins: opportunities, challenges, and applications in WRRFs S. Daneshgar¹, D. Fernandes Del Pozo¹, C. DeGroot², E. Wicklein³, S. Saunders⁴, J. Laurent⁵, J. Climent⁶, G. Bellandi⁷ ¹BIOMATH, Ghent University, ²Western University, ³Carollo Engineers, ⁴Ibis group, ⁵ENGEEES ICube UMR 7357, ⁶Hydrens, ⁷AM-TEAM.
5	Full day	Building blocks towards good modelling practice for N₂O mitigation Part 1: Modelling approaches Part 2: Practical model applications Amanda Lake¹, Mikkel Algren Stokholm-Bjerregaard², Giacomo Bellandi³, Sina Borzooei⁴, Carlos Domingo-Felez⁵, Andreas Froemelt⁶, Sylvie Gillot⁷, Hélène Hauduc⁸, Fabio Polesel⁹, Laura Debel Simonsen¹⁰, Shuting Wang¹¹, Tianyu Lei¹², Benedek Gy. Plósz¹³, Jose Porro¹⁴ ¹Jacobs, ²NIRAS Aalborg, ³AM Team, ⁴IVL Swedish Environmental Research Institute, ⁵Advanced Research Centre, ⁶Eawag, ⁷INRAE Lyon, ⁸DYNAMITA SAS, ⁹DHI A/S, ¹⁰Krüger, Veolia, ¹¹The University of Queensland, ¹²PROSYS, ¹³University of Bath, ¹⁴Cobalt Water Global
6	Full day	Looking into nature-based solutions: bridging mechanistic and machine learning modelling for wastewater treatment Bernhard Pucher¹, Dwight Houweling², Nicolas Forquet³, Joaquim Comas⁴ ¹BOKU University Vienna, ²Polytechnique Montreal, ³INRAE, UR-REVERSAAL, ⁴Catalan Institute for Water Research, University of Girona
7	Full day	Aplicaciones prácticas de modelos al rediseño y operación de EDAR: retos y oportunidades (Workshop in Spanish)

Tuesday June 16, 2026

Activity and Moderators	Time	Details	Speakers
Opening Session	8:30-9:30	Welcome Remarks and Keynote Speaker	E. Ayesa (CEIT)
Session 1: Digital Twins Moderators: TBD	9:30-11:00	<i>The journey of a full-scale tertiary treatment plant model: metamorphosis from a virtual pilot to a digital twin</i>	Giacomo Bellandi (AM-Team)
		<i>High-frequency parameter updates in digital twins: report on a 2-week challenge test for autonomous adaptive operation</i>	Pooria Ghorbani Bam (University of California Irvine)
	11:00-11:30	Coffee	
Session 2: Resource Recovery modelling Moderators: TBD	11:30-1:00	<i>Agile benchmarking framework for wastewater treatment and resource recovery technologies</i>	Yalin Li (Rutgers, The State University of New Jersey)
		<i>Fermentation model for improved VFA prediction integrating reaction thermodynamics</i>	Eric Rovira (CEIT)
		<i>Plant wide modelling of phosphorus removal with iron precipitation and recovery in the form of vivianite</i>	Mathieu Sperandio (INSA)
	1:00-2:30	Lunch time	
Session 3: Mass Transport modelling Moderators: TBD	2:30-4:00	<i>Measuring what has been missing: floc settling data to challenge current clarifier models</i>	Nam Ngo (DC Water)
		<i>Toward a real-time digital twin: accelerating activated sludge reactor CFD via AI-enhanced hydrodynamic surrogate model</i>	Delia Trifi Rufino (Universitat Jaume I)
	4:00-5:00	Coffee + Poster session	
Workshop Summaries Moderators: TBD	5:00-6:00	WS 1 – WS 4	
Dinner	7:30-9:00	Dinner	

Wednesday June 17, 2026			
Activity and Moderators	Time	Details	Speakers
Opening Session	8:30-9:30	Keynote Speaker	Peter Vanrolleghem
Session 4: N2O modelling Moderators: TBD	9:30-11:00	<i>Improving the accuracy to quantify nitrous oxide emissions from wastewater treatment</i>	Shuting Wang (University of Queensland)
		<i>Archetypal microbiome profiles as indicators of nitrous oxide emission states in activated sludge: a data-driven approach</i>	Cheng Chen (Eawag)
		<i>Do we need multi-population models to reflect nitrification stability and N2O risk in low DO nitrogen removal?</i>	McKenna Farmer (Black & Veatch)
	11:00-11:30	Coffee break	
Session 5: Hybrid modelling Moderators: TBD	11:30-1:00	<i>Hybrid models to identify unknown processes: serial, parallel or embedded?</i>	Florian Wenk (EAWAG)
		<i>Data-driven Identification of sparse and interpretable ASM-dynamics</i>	Antonio Silveira (Ghent University)
	1:00-2:30	Lunch time	
Session 6: Control-based solutions Moderators: TBD	2:30-4:00	<i>A temporal interpretability pipeline for reinforcement learning control in water resource recovery facilities</i>	Cristian Camilo Gómez Cortés (Ghent University)
		<i>Optimizing mainstream partial denitrification-Anammox through advanced control strategies: multi-site demonstration</i>	Manel Garrido (Brown and Caldwell)
	4:00-5:00	Coffee + Poster session	
Workshop Summaries Moderators: TBD	5:00-6:00	WS 5 – WS 8	
	6:30-11:00	Gala Dinner	

Thursday June 18, 2026

Activity and Moderators	Time	Details	Speakers
Session 7: Biofilm granular modelling Moderators: TBD	8:30-10:00	<i>A new biofilm model for enhanced simulation of wastewater treatment processes</i>	Kepa Olaciregui Arizmendi (Ceit-BRTA)
		<i>Modelling nitrification in densified activated sludge systems requires to consider a dynamic KO_2 for granular autotrophs</i>	Mathilde Lepage (INRAE REVERSAAL - SUEZ International)
		<i>Modeling granule and floc dynamics in full-scale densified activated sludge system</i>	Wendell Khunjar and Ron Latimer (Metro Water Recovery, Denver, Hazen and Sawyer)
	10:00-10:30	Coffee break	
Session 8: Soft sensors Moderators: TBD	10:30-12:00	<i>Machine Learning Assisted Framework For N_2O Emission Prediction At A Full-scale Sewage Treatment Works In The UK</i>	Xiaobo (Agnes) Shen (University of Surrey)
		<i>Navigating Autocorrelation And Dimensionality In Full-Scale Reverse Osmosis Facilities: An NDMA Modelling Case Study</i>	Kate Newhart (Oregon State University)
Closing Session	12:00-1:00	Closing remarks WRRmod 2028 Outlook	
	1:00-2:30	Lunch, depart	

POSTERS

- A Plant-Wide Modelling And Decision Support Approach For Sustainable WRRF Configuration Selection. Alex Gaona (Universitat Autònoma de Barcelona)
- Anaerobic Dynamic Membrane Bioreactors: Modeling Trade-offs Between Biogas Production And Filtration. Renata Starostka (University of Michigan)
- Chemical Dosing Optimization In A Large Industrial WRRF: Dynamic And CFD Modelling To Boost Implementation. Giacomo Bellandi (AM-Team)
- Computational Fluid Mechanics Based Neural Network Model For Predicting Dissolved Oxygen In A Full-scale Aeration Basin. Petri Urkonnen (Aalto University)
- Coupled Biokinetic-hydrodynamic Model For Antibiotic Removal In A Plug-flow Anaerobic Fixed Bed Reactor. Elias Gabriel (University of São Paulo)
- Data-Driven Framework For Predicting Dynamic Effluent Solids Of Full-Scale HiCS: Addressing Data Sparsity And Imbalance. Ahmed AlSayed (Western University)
- Digital Support Tools For Net-Zero In Industrial Wastewater Treatment: Coupling Plant-Wide Model With Carbon Accounting. Tianyu Lei (Technical University Of Denmark)
- Digital Twins For N-Removal: Integrating Hybrid Modelling, MPC, Soft Sensor, And Day-Ahead Pricing In Utility Operations. Wouter Maenhout (Aquafin NV)
- Dynamics Of Granules And Floc Inventory In Full Scale Municipal WWTP: Model Response, Gaps And Future Needs. Farhang Forouzan Mehr (Suez)
- Eleven Full-Scale Projects In Five Years: Navigating The Journey Of Using Hybrid Digital Twins For Operations And Design. Cheng Yang (Jacobs Engineering Group Inc.)
- Extending Death Regeneration Model In Accurate Prediction Of Dynamic Biogas Production From Wastewater Resource Recovery. Güçlü Insel (Istanbul Technical University)
- Flow Equalisation At Käppala Wastewater Treatment Plant Using Weather Forecasts And Model Predictive Control. Hanna Molin (IVL / Lund University)
- Identification And Modelling Of Key Metabolic Fluxes Correlated To Biodegradation Of Organic Micropollutants. Alen Simonic (Technical University of Denmark)
- Knowledge-informed, Data-driven Framework For Long-term Inlet Flow Forecasting. Jingyu Ge (The University of Queensland)
- Leveraging Simulators To Innovate Operator Training: City Of Roseville, California SimuWorks Platform Development. Jacob Barclay (Hatch)
- Model Informed Piloting, Startup, And Knowledge Capture Of Mobile Carrier Activated Sludge Systems. Katie Printz (Jacobs)
- Model Predictive Control For N₂O Mitigation In Full-Scale Wastewater Treatment Plants. Laura Debel (Veolia)
- Numerical Investigation Of Hydrocyclone Performance For Wastewater Applications Using Drift-Flux Modelling. Jyothis Arumughan (Western University)
- O₂ And N₂O Gas Mass-transfer Efficiency Associates With Microbial Communities Driving Compact Floc Solid Fractions. Benedek Plosz (Oslo Metropolitan University)

- Operationalizing Data-Driven Forecasting To Reduce Energy, Cost And Emissions. Jiawei Liu (The University of Michigan)
- Perspectives For Integrating Phosphorus Recovery In The Manresa WWTP: A Preliminary Plant-wide Modelling Study. Alex Gaona (Universitat Autònoma de Barcelona)
- Predicting R-Floc And Clarifier Performance Using Routine Process Data: Autocalibration And Data-Driven Approaches. Patrick Dunlap (Black & Veatch)
- Prioritizing Experiments On N₂O Reduction Dynamics With Explainable Machine Learning. Laurence Strubbe (Eawag)
- Quaternary Treatments For Micropollutants Removal In WWTPs: A Techno-economic Modelling Approach. Sara Piraldi (Politecnico di Milano)
- Real-time Urban Odour Dispersion Mapping With A Deep Learning Surrogate Model Trained On CFD. Aina Macias (Universitat Jaume I)
- Reduced Order Models For Plant Operation - DARROW Project Case Study. César Arenas (Ceit - University of Navarra)
- Revisiting Extended Kalman Filters As Soft Sensors For Enhanced Monitorization Of Nitrification And Denitrification In R. Beñat Elduayen (CEIT)
- Scaling Bio-Kinetic Simulations For Reinforcement Learning: A Framework For APC Development In WWTPs. Jeremy Versteegh (Hoogheemraadschap Hollands Noorderkwartier)
- Simulating The Diffusion Of Distributed Water Reuse Systems: An Agent-Based Approach For Barcelona. Pol Vidal Lamolla (Universidad de Valladolid)
- Spatial Gas Hold-Up Correlations To Improve Gas-Liquid Transfer Modeling In BAFs. Ahlem Filali (INRAE-PROSE)
- Sustainability Assessment Of Biological And Physical-chemical Systems For FGD Using CRITIC-TOPSIS Based DST. Yizhen Wang (Universitat Autònoma de Barcelona)
- Toward Energy-Efficient Facilities: Modeling Aeration Savings From Primary Filtration Coupling With AGS. Derya Dursun Balci (Caliskaner Water Technologies)
- Towards A Framework For Model-based Design Of WRRFs. Erik U. Lindblom (IVL Swedish Environmental Research Institute)
- WaterRAG: An Agentic RAG To Support Water Industry Transitions To Net-Zero. Mudi Zhai, Haoran Duan (University of New South Wales)
- When Do Machine Learning Model Explanations Reflect Real Process Behavior? A Case Study On MP Removal With GAC. Nicolas Neuenhofer (Eawag)
- Zero-dimensional And One-dimensional Modeling Of Hybrid Densified Sludge Process. Jin Wang (Toulouse Biotechnology Institute)